



USER GUIDE

HIAS

Hall Invigilator Allocation System

St. Xavier's Catholic College of Engineering (Autonomous), Nagercoil

Version **1.1.0** · Windows desktop application · Works fully offline

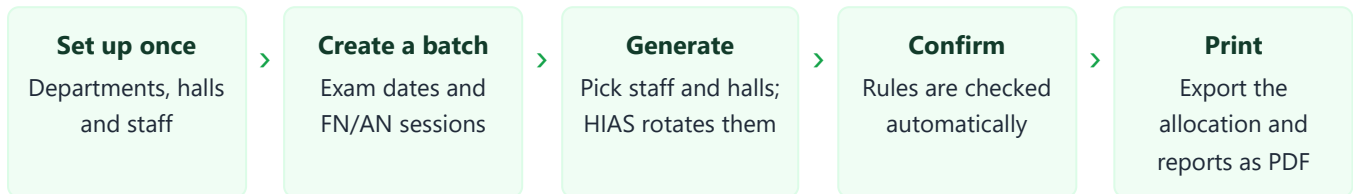
Developed by **Anushath S, Akhil Koska A** and **Harish S**

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1. What HIAS does

HIAS assigns invigilators (staff) to examination halls for every exam session, and rotates those assignments fairly. It remembers who has already been in which hall, so nobody is sent to the same hall again until they have had a turn in every other hall of the rotation.



- **Fully offline.** HIAS needs no internet connection and no server. Everything is stored on the computer it is installed on.
- **No login.** The app opens straight to the dashboard; there are no accounts or passwords inside HIAS. Anyone who can use this computer's Windows account can use HIAS, so the protection is the Windows account and physical access to the computer: keep HIAS on a computer (and Windows account) that only authorised people use.
- **Automatic rule checks.** Before an allocation is confirmed, HIAS checks that no hall or person is double-booked, nobody repeats a hall too early, and inactive staff or halls are not used.

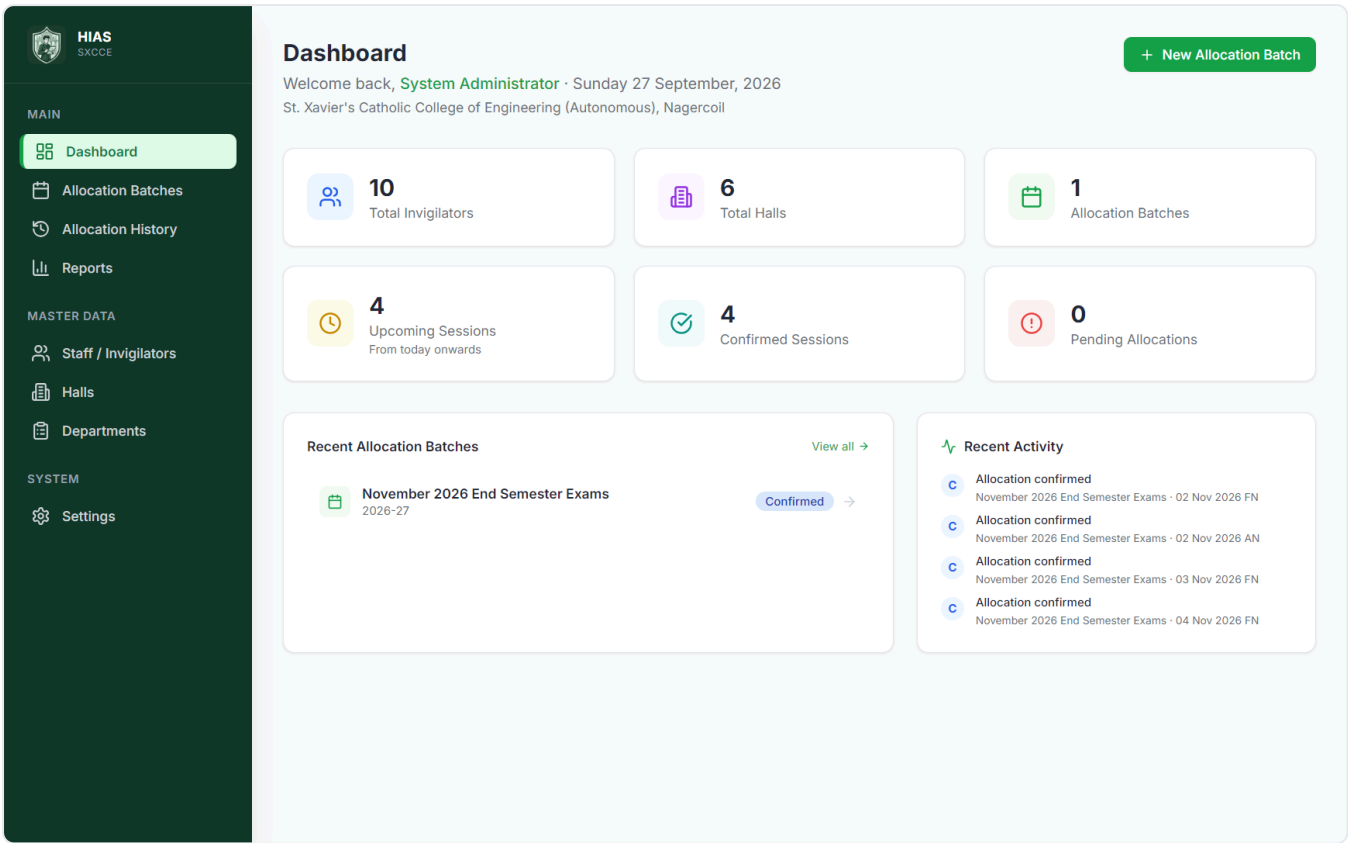
2. Installing HIAS

- 1 Copy `HIAS-Setup-1.1.0.exe` (about 76 MB) to the computer and double-click it.
- 2 If Windows shows "*Windows protected your PC*", click **More info** and then **Run anyway**. This appears because the installer is not code-signed; it is safe.
- 3 When Windows asks "*Do you want to allow this app to make changes to your device?*", click **Yes**. HIAS installs for everyone who uses the computer, which needs administrator permission once.
- 4 Choose the installation folder (the default is fine) and click **Install**.
- 5 Leave **Run HIAS** ticked and click **Finish**. Next time, open HIAS from the **HIAS** shortcut on the desktop or in the Start menu.

Installing a newer version over an older one upgrades it and keeps all data. To remove HIAS, use Windows **Settings** → **Apps** → **Installed apps** → **HIAS - Hall Invigilator Allocation System** → **Uninstall**.

Where is my data? All data is kept in one file on this computer: `%APPDATA%\eias\eias.db`. Updating or reinstalling HIAS keeps this file. Uninstalling the app does not delete it either. See section 9 for backups.

3. A tour of the screen



The dashboard. The sidebar on the left is always available. (All names in this guide are sample data.)

Sidebar item	What it is for
Dashboard	Totals at a glance: invigilators, halls, batches, upcoming and confirmed sessions, and sessions still pending allocation. Click a card to jump to it.
Allocation Batches	Create an exam batch (for example "November 2026 End Semester Exams") and open it to allocate each session.
Allocation History	Search every confirmed allocation by batch, session, staff member or hall.
Reports	Staff-wise, date-wise, timetable and audit reports, each exportable as PDF.
Staff / Halls / Departments	The master data HIAS allocates from.
Settings	College name, default session timings, backup and restore, and rotation restart.

A **Back** link at the top of each page returns to the previous screen. The button at the top of the sidebar shrinks it to icons (hover over an icon to see its name); HIAS remembers your choice. In a narrow window, for example when HIAS fills half the screen, the sidebar shows icons automatically. The bottom of the sidebar reads **Offline • Local database**: everything you see is stored on this computer.

4. First-time setup

Do these steps once, in this order. You can come back and change anything later.

4.1 College profile and session timings

Open **Settings** → **College Profile**. The **College Full Name** is printed at the top of every PDF; the **Short Name** appears under the HIAS logo in the sidebar. Click **Save**.

Under **Settings** → **Session Timings**, set the usual reporting, start and end times for the Forenoon (FN) and Afternoon (AN) sessions. New batches start with these times; you can still change them per session.

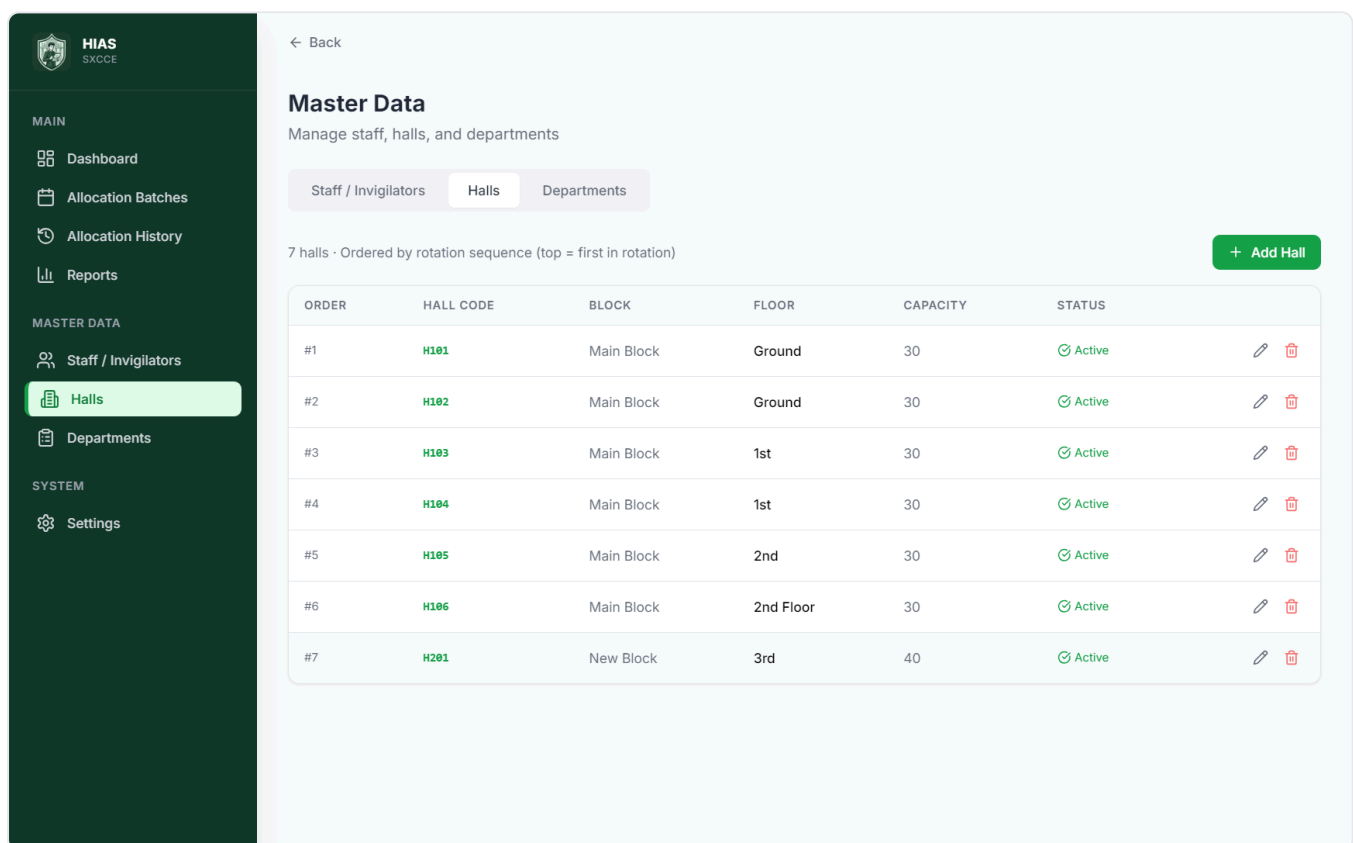
4.2 Departments

Open **Departments** and add each department with a short **Code** (for example **CSE**) and a full **Name**.

Reports use the short code. The code is saved exactly as you type it, but capital and small letters count as the same code: once **CSE** exists, **cse** or **Cse** is refused as a duplicate. The bin icon removes a department that no staff member belongs to, after asking you to confirm.

4.3 Halls

Open **Halls** → **Add Hall**. Only the **Hall Code** (for example **H101**) is required. **Block** and **Floor** (Ground, 1st, 2nd...) say where the hall is; they are printed next to the hall code on the allocation sheet so invigilators know where to go. Capacity is optional. Untick **Active** for a hall that is temporarily unavailable; inactive halls are never allocated.



The screenshot shows the HIAS application interface. On the left is a dark green sidebar with the HIAS SXCE logo and a menu with categories: MAIN (Dashboard, Allocation Batches, Allocation History, Reports), MASTER DATA (Staff / Invigilators, Halls, Departments), and SYSTEM (Settings). The 'Halls' option is highlighted. The main content area has a 'Back' link and a 'Master Data' section with the subtitle 'Manage staff, halls, and departments'. Below this are tabs for 'Staff / Invigilators', 'Halls', and 'Departments', with 'Halls' selected. A status bar indicates '7 halls · Ordered by rotation sequence (top = first in rotation)' and a '+ Add Hall' button. A table lists the halls with columns: ORDER, HALL CODE, BLOCK, FLOOR, CAPACITY, STATUS, and action icons (edit and delete). All halls are currently 'Active'.

ORDER	HALL CODE	BLOCK	FLOOR	CAPACITY	STATUS	
#1	H101	Main Block	Ground	30	Active	 
#2	H102	Main Block	Ground	30	Active	 
#3	H103	Main Block	1st	30	Active	 
#4	H104	Main Block	1st	30	Active	 
#5	H105	Main Block	2nd	30	Active	 
#6	H106	Main Block	2nd Floor	30	Active	 
#7	H201	New Block	3rd	40	Active	 

Halls. The pencil edits a hall; the bin removes a hall that has never been allocated, after asking you to confirm (for a used hall, untick Active instead).

4.4 Staff (invigilators)

Add staff one at a time with **Add Staff**, or many at once with **Import Excel**. The file **HIAS-Staff-Import-Template.xlsx** , supplied with HIAS, has the right headings ready. For an import, the first row of the sheet

must contain these column headings:

Column	Required?	Notes
Staff ID	Yes	Must be unique. Rows whose Staff ID already exists are skipped.
Name	Yes	Full name of the staff member.
Department	Yes	The department's code or full name; it must already exist in Departments.
Designation, Email	No	Optional details.

After the import, HIAS shows how many staff were added and lists each skipped row with the reason (for example "Row 5: Department CIVIL does not match any existing department"), so you can correct the sheet and import it again.

The screenshot shows the HIAS SACOE Master Data interface. On the left is a dark green sidebar with navigation links: MAIN (Dashboard, Allocation Batches, Allocation History, Reports) and MASTER DATA (Staff / Invigilators, Halls, Departments). The main content area is titled 'Master Data' with the subtitle 'Manage staff, halls, and departments'. It features three tabs: 'Staff / Invigilators' (selected), 'Halls', and 'Departments'. Below the tabs, it says '10 staff members' and has buttons for 'Import Excel' and '+ Add Staff'. A table lists 10 staff members with columns: STAFF ID, NAME, DEPARTMENT, DESIGNATION, and STATUS. Each row includes a green 'Active' status with a checkmark and edit/delete icons.

STAFF ID	NAME	DEPARTMENT	DESIGNATION	STATUS
STF001	Anitha R	Computer Science and Engineering	Assistant Professor	Active
STF004	Dinesh K	Computer Science and Engineering	Assistant Professor	Active
STF007	Gokul P	Computer Science and Engineering	Assistant Professor	Active
STF101	Imran K	Computer Science and Engineering	Associate Professor	Active
STF002	Bala S	Electronics and Communication Engineering	Assistant Professor	Active
STF005	Esther J	Electronics and Communication Engineering	Assistant Professor	Active
STF008	Hema V	Electronics and Communication Engineering	Assistant Professor	Active
STF102	Janani L	Electronics and Communication Engineering	Assistant Professor	Active
STF003	Chitra M	Mechanical Engineering	Assistant Professor	Active
STF006	Faizal A	Mechanical Engineering	Assistant Professor	Active

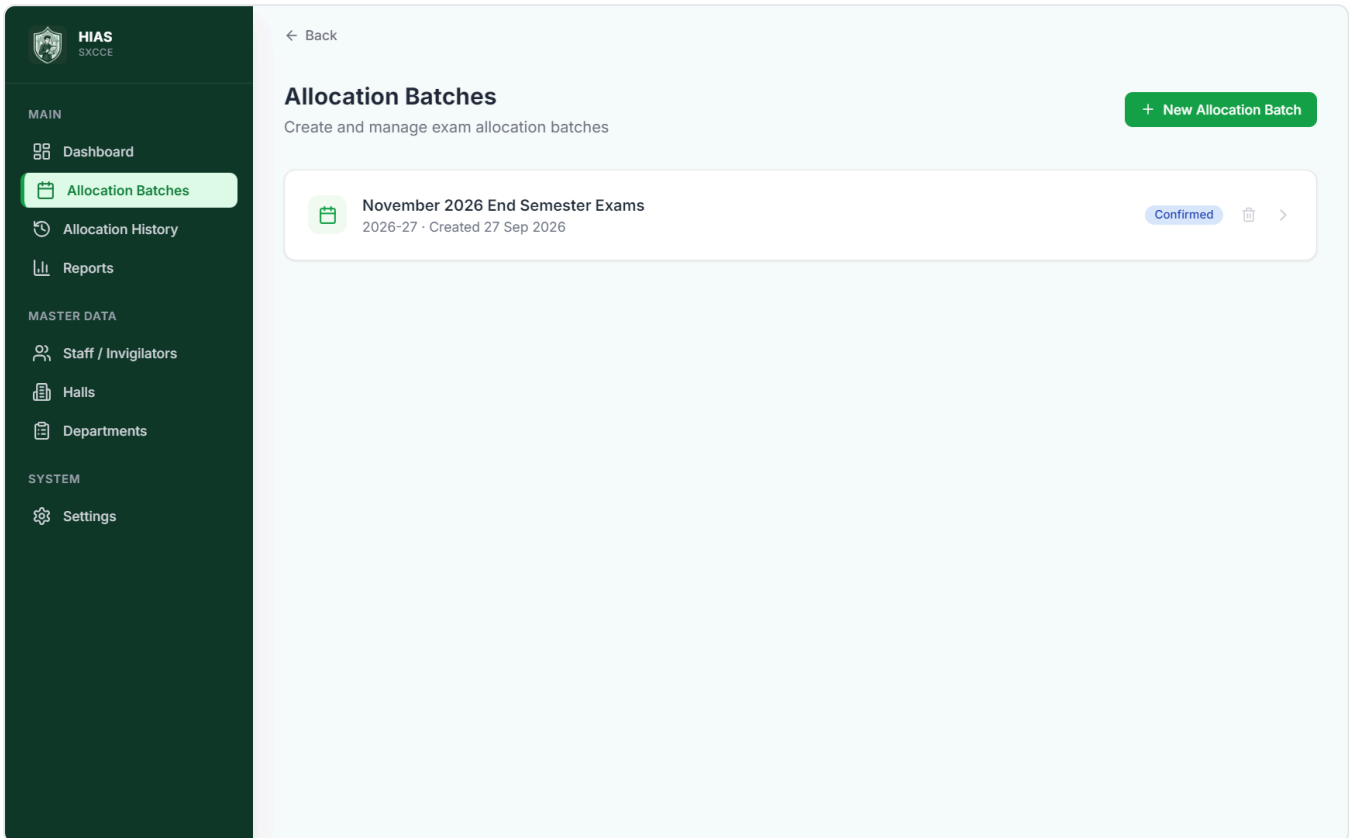
Staff / Invigilators, with Import Excel and Add Staff at the top right.

Deactivate or delete? The bin icon first **deactivates** a staff member: they stay in past records but are no longer allocated. An inactive staff member with no allocation history can then be **deleted permanently**; HIAS asks you to type `DELETE` to confirm.

5. Creating an allocation batch

A **batch** is one examination series, such as an end-semester or internal assessment exam. Open **Allocation Batches** → **New Allocation Batch**. The wizard has three steps:

- 1 **Batch details.** Enter the batch name, the academic year (for example 2026-27) and the number of exam days.
- 2 **Pick the dates** on the calendar. Sundays and public holidays are highlighted so you do not pick them by mistake.
- 3 **Sessions and timings.** For each date, choose Forenoon (FN), Afternoon (AN) or both. Timings come from Settings; expand a date to change its times.



Allocation Batches. A batch shows **Draft** until every session in it is confirmed, then **Confirmed**.

To delete a whole batch, click its bin icon and type **DELETE** . This removes all its sessions, allocations and rotation records and cannot be undone.

6. Allocating invigilators to a session

Open a batch to reach its **Allocation Workspace**. Each session appears as a tab (for example *02 Nov 2026 FN*).

- 1 Select a session tab and click **Select Staff & Generate**.
- 2 Choose the staff who are on duty and the halls in use for this session. **The number of staff must equal the number of halls**; HIAS tells you if they differ. You can:
 - tick people one by one, or use **Select all** for a department;
 - type part of a **name or Staff ID** in the search box to find someone quickly, and part of a **hall code or block** in the halls search box to find halls the same way;
 - click **Upload Excel** to select everyone on a duty list at once (see below);

- click **Copy previous session**, or pick a session under **Copy from a session...**, to start from the staff and halls ticked in an earlier session (see below).

- 3 Click **Generate Allocation**. HIAS assigns everyone at once: each person gets a hall they have not yet had in their current round of halls, drawn at random rather than in a fixed order, and halls are spread evenly over time. Generating the same session again with the same staff and halls gives the same result.
- 4 Review the list. To change an assignment, click the pencil on that row and choose a hall: the invigilator of that hall takes this person's hall, so the two **swap**. HIAS only offers swaps that keep the rules satisfied for **both** people; the other halls are marked *Current*, *In cycle* or *Swap blocked* (point at one to see why). **Regenerate** starts the session over; to start again from the same staff and halls, choose **This session (current draft)** under **Copy from a session....**
- 5 Click **Confirm Allocation**. HIAS runs all rule checks; if something is wrong it explains what, and nothing is saved until it is fixed.
- 6 Click **Export PDF** to save the printable allocation sheet for this session.

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November 2026 End Semester Exams

2026-27 · Allocation Workspace

02 Nov 2026 FN • 02 Nov 2026 AN • 03 Nov 2026 FN • 04 Nov 2026 FN • [+ Add Session](#)

Date: Mon, 02 Nov 2026 Session: Forenoon (FN) Report: 09:30 AM Exam: 10:00 AM – 01:00 PM [Confirmed](#)

[Edit Schedule & Timings](#) [Export PDF](#)

#	STAFF ID	NAME	DEPARTMENT	ASSIGNED HALL	TYPE
1	STF001	Anitha R	CSE	H101	Auto-generated
2	STF004	Dinesh K	CSE	H104	Auto-generated
3	STF002	Bala S	ECE	H102	Auto-generated
4	STF005	Esther J	ECE	H105	Auto-generated
5	STF003	Chitra M	MECH	H103	Auto-generated
6	STF006	Faizal A	MECH	H106	Auto-generated

6 invigilators · 0 admin-edited

A confirmed session. Rows you changed by hand are marked "Admin Edited" instead of "Auto-generated" (in a swap, both people's rows); swapping back to the generated halls removes the mark.

Uploading a duty list from Excel

Instead of ticking people one by one, upload an Excel (`.xlsx` , `.xls`) or `.csv` file listing the invigilators on duty:

- 1 In the selection window, click **Download template**. It is an Excel sheet of all active staff with the columns **Staff ID**, **Name**, **Department** and **Hall**.

- 2 Delete the rows of people who are not on duty. Optionally type a hall code (for example H101) in the Hall column to choose which halls are open. Save the file.
- 3 Click **Upload Excel** and choose the file. Everyone listed is selected (and the halls, if a Hall column is filled in); anything already ticked is replaced.
- 4 Check the summary. Any row that could not be used is listed with the reason, for example an unknown Staff ID, a name shared by two people, or a person listed twice.

Your own duty list works too, as long as its first row has a **Staff ID** column (or a **Name** column). A sheet with no headings is read as a list of Staff IDs in the first column. The Hall column only decides **which halls are used**; HIAS still decides who goes to which hall, so the rotation stays fair.

Copying the staff and halls of another session

When the people on duty barely change from one session to the next, start from an earlier session's ticks instead of ticking everyone again:

- **Copy previous session** ticks the same staff and halls as the nearest earlier session of this batch (for the first session of a batch, the latest earlier session of any batch).
- **Copy from a session...** lists every session that has an allocation, this batch first and newest first. After **Regenerate** it also offers **This session (current draft)**.

The copy replaces what is ticked; then add or remove people as usual. Staff and halls that are no longer active are left out, and the summary says how many. Only the **selection** is copied, never who had which hall: **Generate Allocation** draws everyone's hall afresh, exactly as if you had ticked them by hand.

Why confirming matters. The rotation only moves forward when a session is **confirmed**, and HIAS confirms sessions strictly in order: a session can be confirmed only when every earlier session of the batch is confirmed. Sessions you add or reschedule later take their place in date order among the sessions not yet confirmed; confirmed sessions never move.

The rules HIAS checks

Rule	Meaning
R1	A staff member does not return to a hall until they have been through the other halls in the rotation. When the people on duty change a lot, this is occasionally impossible for everyone; HIAS then keeps repeats to the minimum and shows them as yellow "Unavoidable repeat" warnings, which do not block confirmation.
R2	A hall gets only one invigilator per session.
R3	A staff member is in only one hall per session.
R4	Every earlier session of the batch must be confirmed first.
R5 / R6	Inactive staff and inactive halls are never used.
R7	The number of staff equals the number of halls.

7. Reports

Open **Reports**, choose a report type, choose the filters, click **Generate** to preview it, then **Export PDF**. Each PDF carries the college name and the time it was generated. When you export, Windows asks where to save the file (the Downloads folder is suggested).

Report	Use it for	Filters
Staff-Wise Duty Report	Every duty of each staff member: date, session, hall and timings.	Optional: one staff member
Date-Wise Allocation Sheet	The hall-by-hall list for one session, for the notice board. The title names the date and session.	Batch and session
Complete Timetable Grid	One page showing every staff member against every session of a batch.	Batch
Rotation Audit Report	Which assignments were generated automatically and which were changed by an admin (swaps).	Batch

MAIN

Dashboard

Allocation Batches

Allocation History

Reports

MASTER DATA

Staff / Invigilators

Halls

Departments

SYSTEM

Settings

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Reports

Generate and export allocation reports with your college letterhead

Staff-Wise Duty Report
All sessions per staff member

Date-Wise Allocation Sheet
Hall-by-hall seating for posting

Complete Timetable Grid
Full Staff × Session matrix

Rotation Audit Report
Auto vs admin-edited assignments

Allocation Batch *

Session *

Generate

Export PDF

HALL	HALL NAME	CAP	STAFF ID	NAME	DEPT	DESIGNATION	REPORT	EXAM
H101	Main Block Room 101	30	STF001	Anitha R	CSE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H102	Main Block Room 102	30	STF002	Bala S	ECE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H103	Main Block Room 103	30	STF003	Chitra M	MECH	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H104	Main Block Room 104	30	STF004	Dinesh K	CSE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H105	Main Block Room 105	30	STF005	Esther J	ECE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H106	Main Block Room 106	30	STF006	Faizal A	MECH	Assistant Professor	09:30 AM	10:00 AM–01:00 PM

6 records

Date-Wise Allocation Sheet preview, ready to export.

8. Allocation history

Allocation History lets you look up any past confirmed allocation. Filter by batch, session, staff member or hall and click **Search Records**. This is the quickest way to answer "Where was this person on that day?".

HIAS – Hall Invigilator Allocation System

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9. Backup and restore

HIAS keeps its data only on this computer. If the computer fails and there is no backup, the data cannot be recovered. Take a backup after every allocation batch and keep it on a pen drive or another computer.

To take a backup

- 1 Open **Settings** → **Backup & Restore** and click **Backup Database**.
- 2 Choose where to save the file (for example a pen drive). The file is named `hias-backup.db` by default; adding the date to the name helps.

To restore a backup

- 1 Open **Settings** → **Backup & Restore** and click **Restore Database**.
- 2 Choose the backup file and confirm. HIAS first checks that the file really is a HIAS backup (if not, it says so and changes nothing), saves a safety copy of the current data in the data folder (named like `eias.db.before-restore-20261101-093000.bak`, with the date and time), and then replaces it.
- 3 HIAS restarts by itself a moment later and opens with the restored data.

The same steps move HIAS to a new computer: back up on the old computer, install HIAS on the new one, and restore. Everything (staff, halls, batches, allocations and the rotation) travels in that one file. To go back to the data from before a restore, restore the safety copy the same way.

10. Restarting the rotation

Settings → **Restart Rotation** clears the whole rotation history, so every staff member starts a fresh round with every hall open to them again. Use it only at a genuine new start, such as the beginning of an academic year. You must type `RESTART` to confirm. Past allocations and reports are not deleted; only the rotation memory is cleared.

11. Troubleshooting and FAQ

Problem	What to do
"Staff count must equal hall count"	Select the same number of staff and halls for the session.
A staff member or hall is missing from the selection list	Check it is marked Active in Staff or Halls.
Confirm is refused with a rule message (R1–R7)	Read the message; it names the person or hall. Edit that row, or regenerate, then confirm again.
Windows says "Windows protected your PC" when installing	Click More info → Run anyway . The installer is not code-signed.
The app shows an error box at start-up	Note the message, then restore the latest backup (section 9). The message is shown because a silent failure would give no clue.
Can two computers share the same data?	No. Each installation has its own data. Use one computer for allocation, and move data with backup and restore if needed.
Does HIAS need the internet?	No. It works completely offline: Wi-Fi and network cables can be disconnected. Reports and PDFs are made on this computer.
Is there a login or password?	No. HIAS is meant for one trusted computer; whoever can use its Windows account can use HIAS. Lock the computer (Windows key + L) when you leave it.
The data file was deleted or the computer was replaced	HIAS starts with a new, empty database. Restore your latest backup (section 9) to get everything back. Without a backup the data cannot be recovered, which is why regular backups matter.